

Institute of Computer Technology

B. Tech Computer Science and Engineering

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Semester: 6 | Enrollment No: 23162171020 | Batch: 65

Subject: DMW

Practical: 4

Question:

Provide basic information about the Weka tool (purpose, facilities available, and operations performed).

Answer:

Introduction to Weka

Weka (Waikato Environment for Knowledge Analysis) is an open-source data mining and machine learning tool developed at the University of Waikato, New Zealand. It is written in Java and is widely used for data analysis, predictive modeling, and research purposes.

Weka provides a user-friendly graphical interface and supports various machine learning algorithms for data preprocessing, classification, clustering, regression, association rule mining, and visualization.

Purpose of Weka Tool

Weka is mainly used for:

- Data Mining
- Machine Learning
- Predictive Analysis
- Data Preprocessing
- Pattern Discovery
- Research and Educational purposes

It helps users analyze large datasets and extract meaningful information from them.

Main Facilities Available in Weka

Weka provides the following main interfaces:

1. Explorer
 - Used for data preprocessing and applying algorithms.
 - Most commonly used interface.

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2. Experimenter
 - Used to compare the performance of different algorithms.
3. Knowledge Flow
 - Provides a graphical way to design data mining processes.
4. Simple CLI
 - Command Line Interface for advanced users.

Operations that Can Be Performed in Weka

Weka allows the following operations:

1. Data Preprocessing
 - Remove missing values
 - Filter attributes
 - Normalize data
 - Generate random data
2. Classification
 - Decision Trees (J48)
 - Naïve Bayes
 - Random Forest
 - ZeroR, etc.
3. Clustering
 - K-Means
 - EM Algorithm
 - Hierarchical clustering
4. Association Rule Mining
 - Apriori algorithm
5. Regression
 - Linear Regression
 - SMOreg
6. Attribute Selection
 - Select important features from dataset
7. Visualization
 - Graphical representation of data
 - Scatter plots and class distribution graphs

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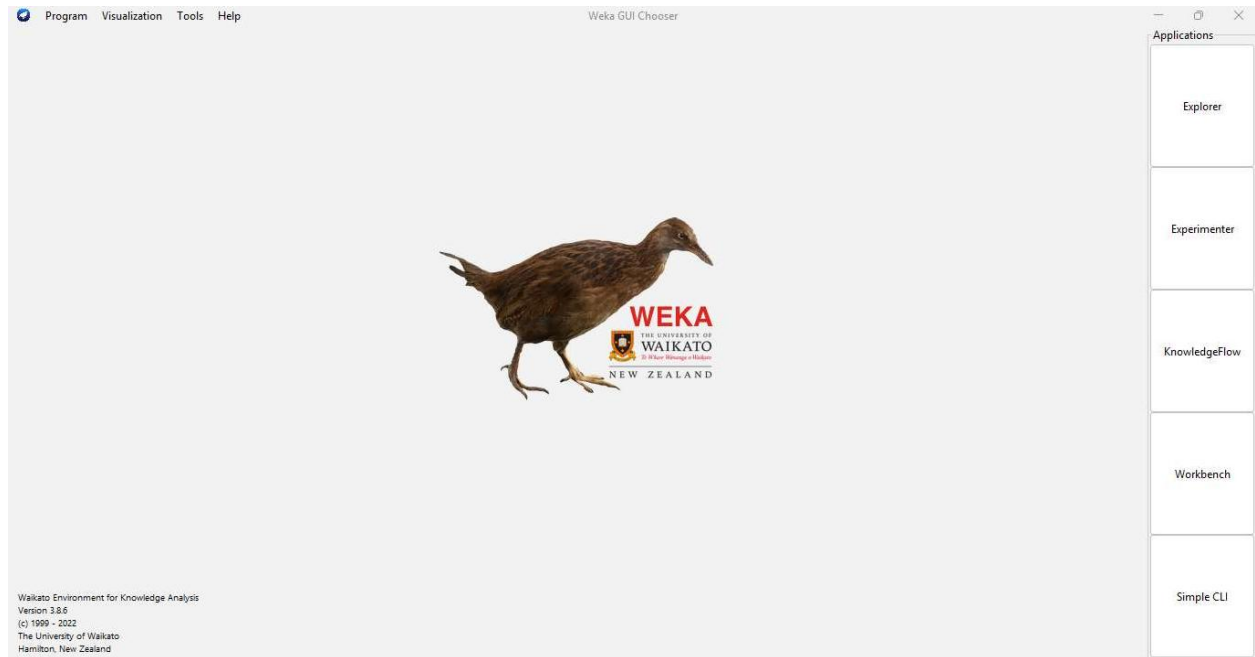
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Weka Tool Exploration:

Steps:

Step-1: Open Weka Tool



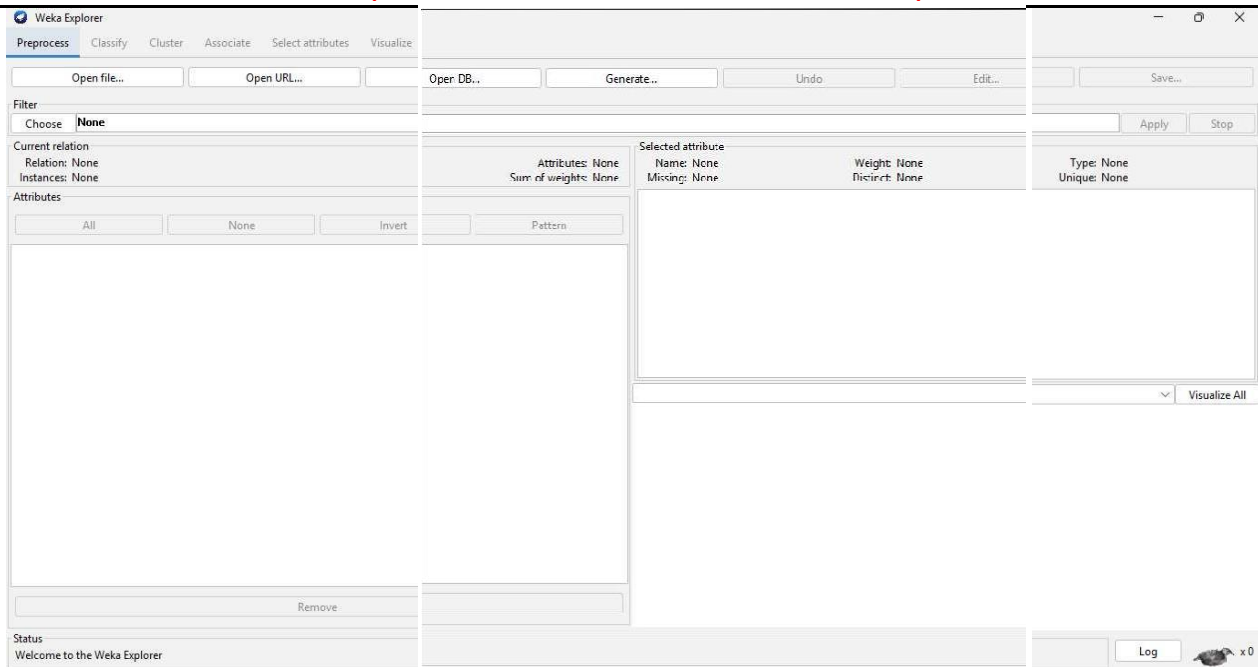
Step-2: Click on Explorer

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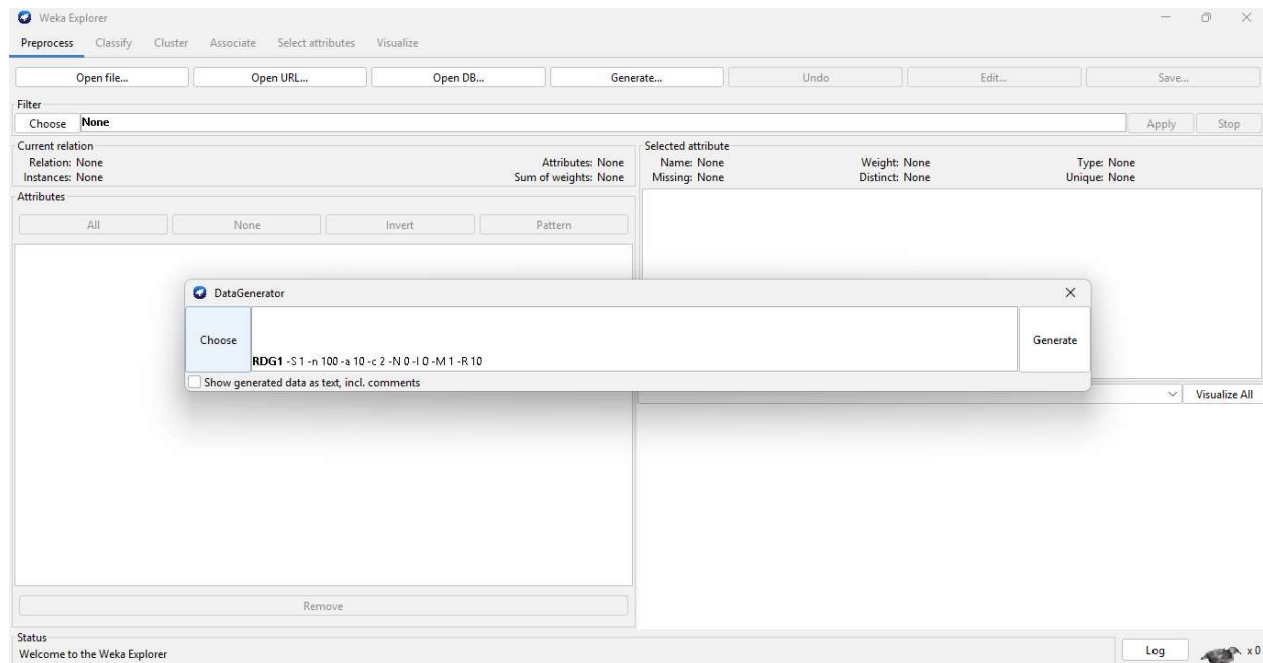
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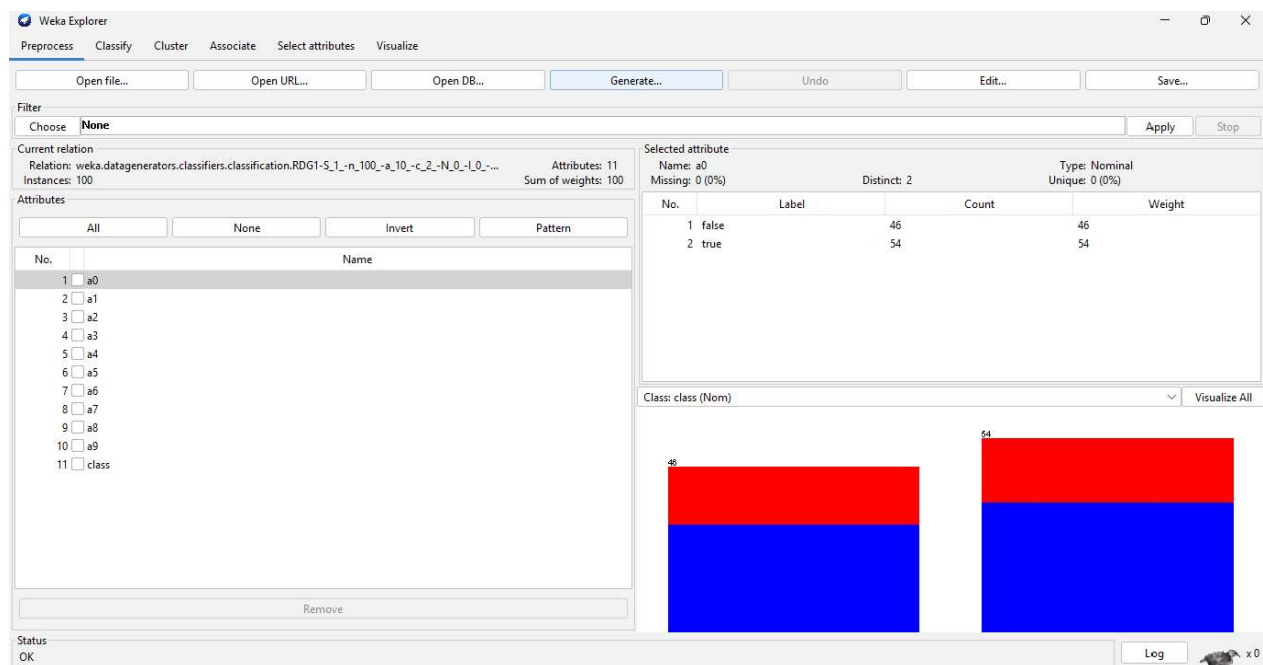
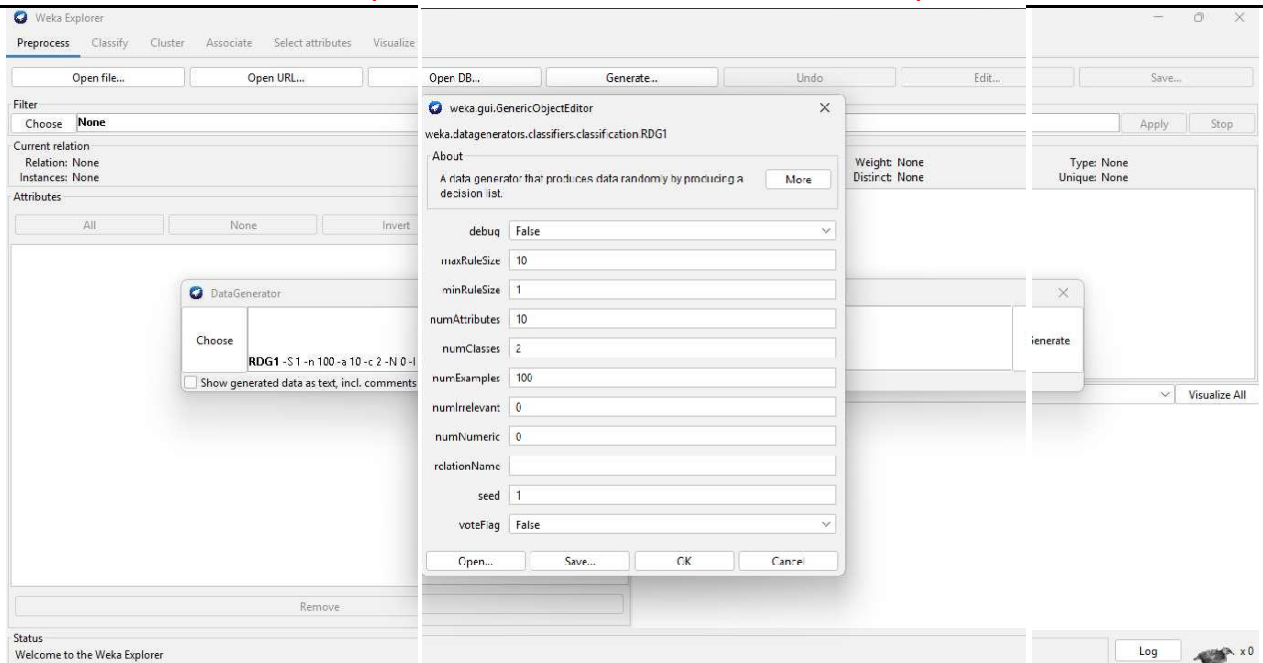
Step-3: Generate Random Data



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Step-4: Classify it

The screenshot shows the Weka Explorer interface with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'ZeroR'. The 'Test options' section shows 'Cross-validation' with 'Folds' set to 10. The 'Result list' on the left shows '08:34:37 - rules.ZeroR' selected. The 'Classifier output' pane displays the following information:

```
==== Classifier model (full training set) ====
ZeroR predicts class value: c0
Time taken to build model: 0 seconds

==== Stratified cross-validation ====
==== Summary ====

Correctly Classified Instances      66      66  %
Incorrectly Classified Instances    34      34  %
Kappa statistic                     0
Mean absolute error                 0.4504
Root mean squared error             0.4743
Relative absolute error             100  %
Root relative squared error         100  %
Total Number of Instances          100

==== Detailed Accuracy By Class ====

          TP Rate  FP Rate  Precision  Recall  F-Measure  MCC  ROC Area  PRC Area  Class
1.000    1.000    0.660    1.000    0.795    ?    0.447    0.638    c0
0.000    0.000    ?        0.000    ?        ?    0.447    0.319    c1
Weighted Avg.    0.660    0.660    ?        0.660    ?        ?    0.447    0.530

==== Confusion Matrix ====

  a b  <-- classified as
66 0 1  a = c0
34 0 1  b = c1
```

Step-5: Cluster it

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'EM -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100'. The 'Cluster mode' section shows 'Use training set' selected. The 'Result list' on the left shows '08:34:48 - EM' selected. The 'Clusterer output' pane displays the following information:

```
Clusterer output
[total] 68.117 35.883
a7
false 37.364 14.636
true 30.753 21.247
[total] 68.117 35.883
a8
false 30.5993 23.4007
true 37.5176 12.4824
[total] 68.117 35.883
a9
false 34.0715 16.9285
true 34.0454 18.9546
[total] 68.117 35.883
class
c0 64.3561 3.6439
c1 3.7609 32.2391
[total] 68.117 35.883

Time taken to build model (full training data) : 0.43 seconds

==== Model and evaluation on training set ====

Clustered Instances

0 68 ( 68%)
1 32 ( 32%)

Log likelihood: -7.28232
```

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(i) What kind of data format is accepted in Weka tool? Provide brief information about default file format with example.

Data Formats Accepted in Weka

Weka supports multiple file formats:

- ARFF (Default and most commonly used)
- CSV (Comma Separated Values)
- TXT
- C4.5 format
- XRFF (XML version of ARFF)

Default File Format: ARFF

The default file format of Weka is **ARFF (Attribute-Relation File Format)**.

ARFF files consist of two main sections:

1. Header Section

- Defines the dataset name
- Defines attributes and their data types

2. Data Section

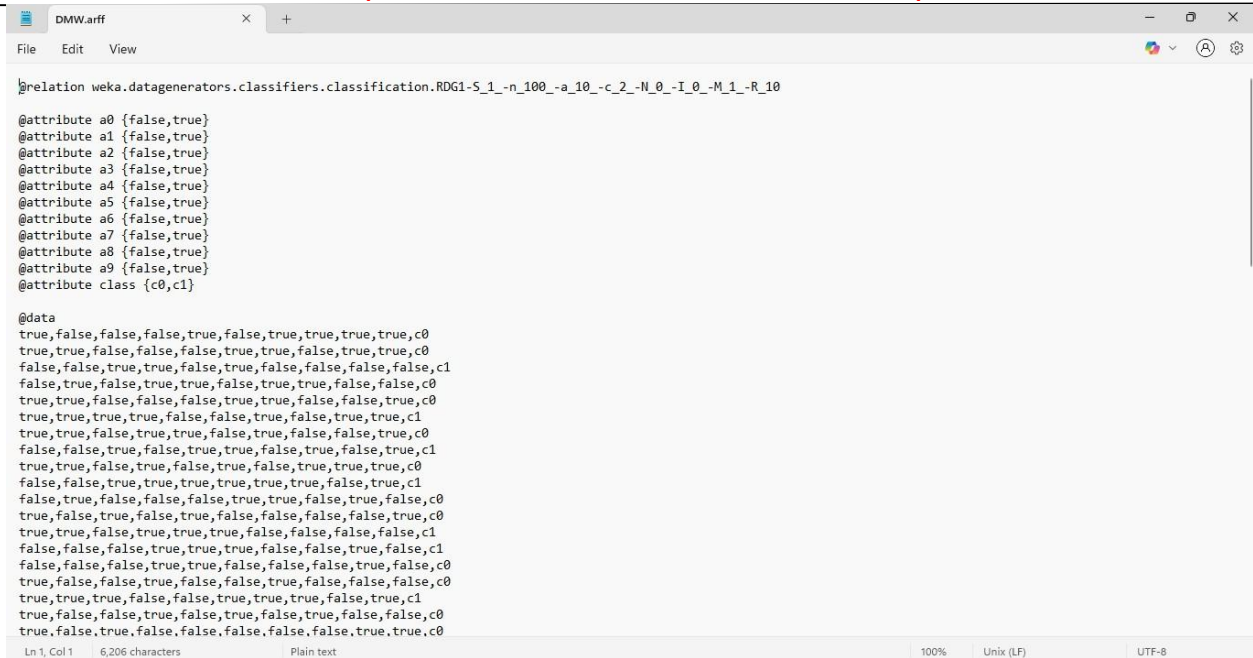
- Contains actual data values

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```
@relation weka.datagenerators.classifiers.classification.RDG1-S_1-n_100-a_10-c_2-N_0-I_0-M_1-R_10

@attribute a0 {false,true}
@attribute a1 {false,true}
@attribute a2 {false,true}
@attribute a3 {false,true}
@attribute a4 {false,true}
@attribute a5 {false,true}
@attribute a6 {false,true}
@attribute a7 {false,true}
@attribute a8 {false,true}
@attribute a9 {false,true}
@attribute class {c0,c1}

@data
true,false,false,false,true,false,true,true,true,true,c0
true,true,false,false,false,true,true,false,true,true,c0
false,false,true,true,false,true,false,false,false,false,c1
false,true,false,true,true,false,true,true,false,false,c0
true,true,false,false,false,true,true,false,false,true,c0
true,true,true,true,false,false,true,false,true,true,c1
true,true,false,true,true,false,true,false,false,true,c1
false,false,true,false,true,true,false,true,false,true,c1
true,true,false,true,false,true,false,true,true,true,c0
false,false,true,true,true,true,true,true,false,true,c1
false,true,false,false,false,true,true,false,true,false,c0
true,false,true,false,true,false,false,false,true,c0
true,true,false,true,true,true,false,false,false,false,c1
false,false,false,true,true,true,false,false,true,false,c1
false,false,false,true,true,false,false,false,true,false,c0
true,false,false,true,false,false,true,false,false,false,c0
true,true,true,false,false,true,true,true,false,true,c1
true,false,false,true,false,true,false,true,false,false,c0
true,false,true,false,false,false,false,true,true,c0
```

Explanation:

- @relation: Name of dataset
- @attribute: Defines attributes and their types
- @data: Actual dataset values

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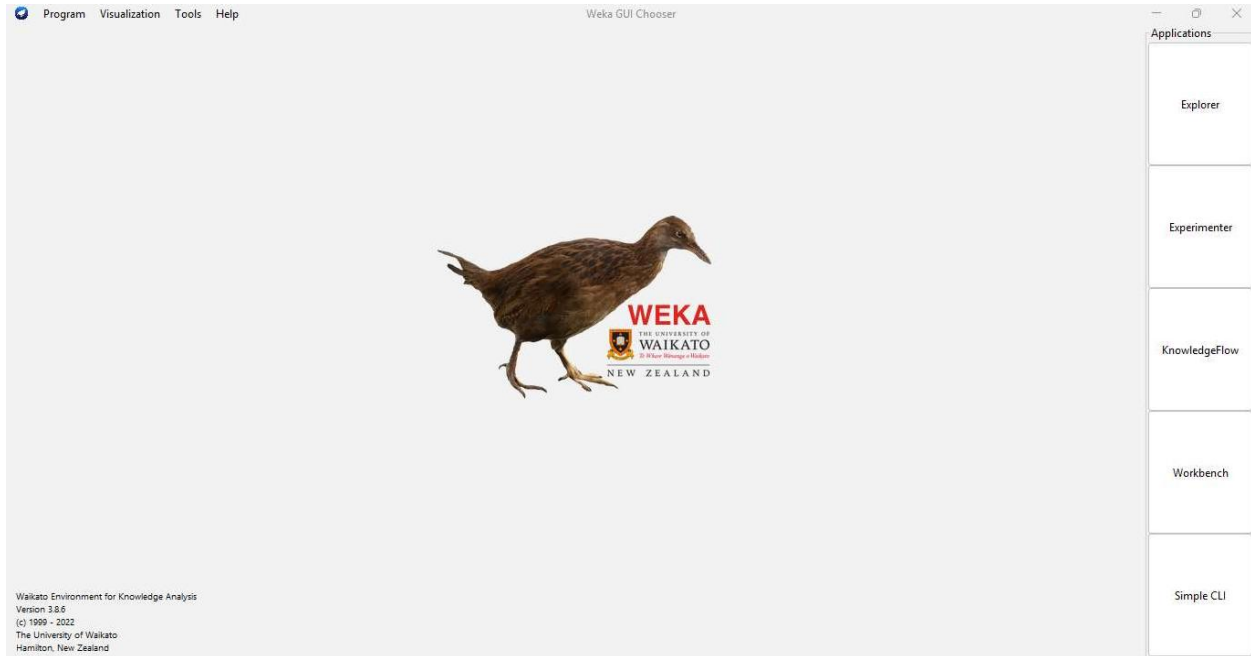
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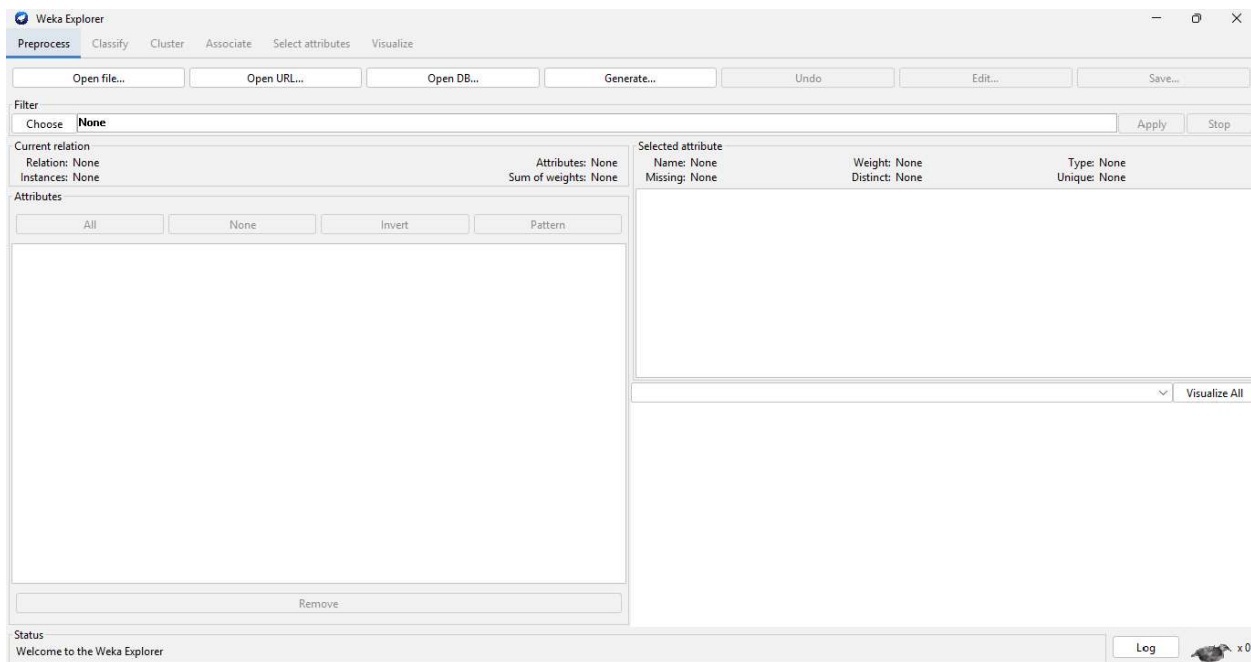
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(ii) Demonstrate how to convert the csv file into the file format which is accepted by Weka.

Step-1: Open the Weka Tool



Step-2: Click on Explorer



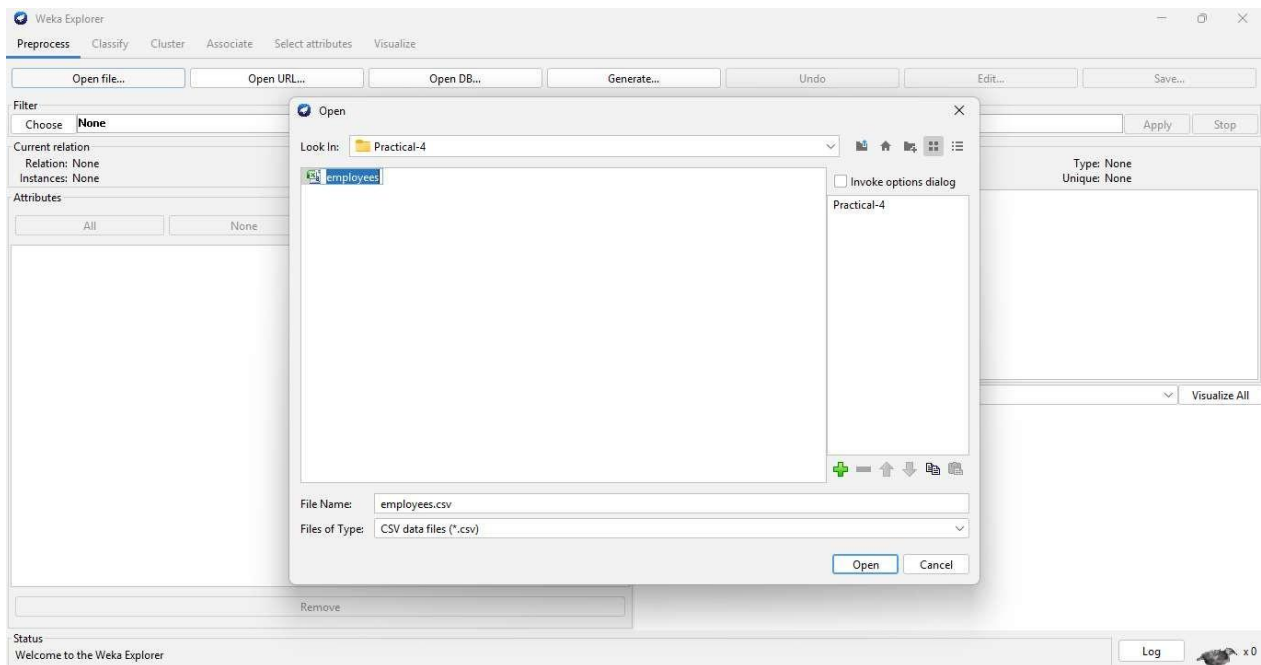
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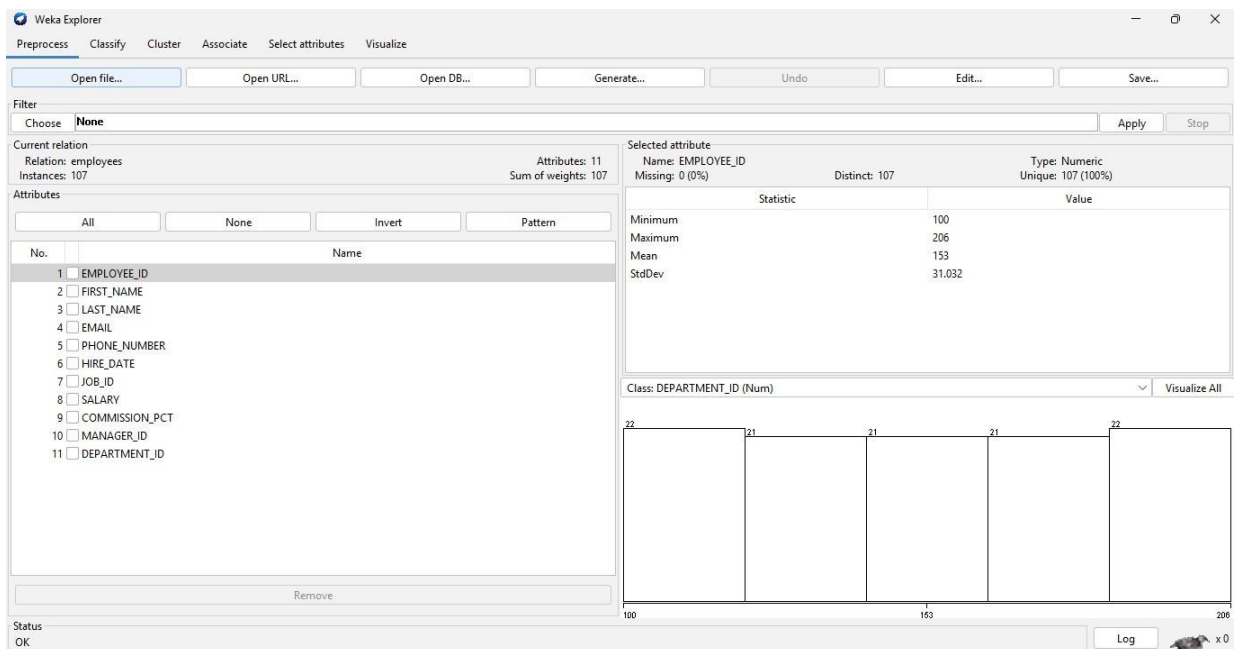
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Step-3: Click on Open file



Step-4: Open the CSV file



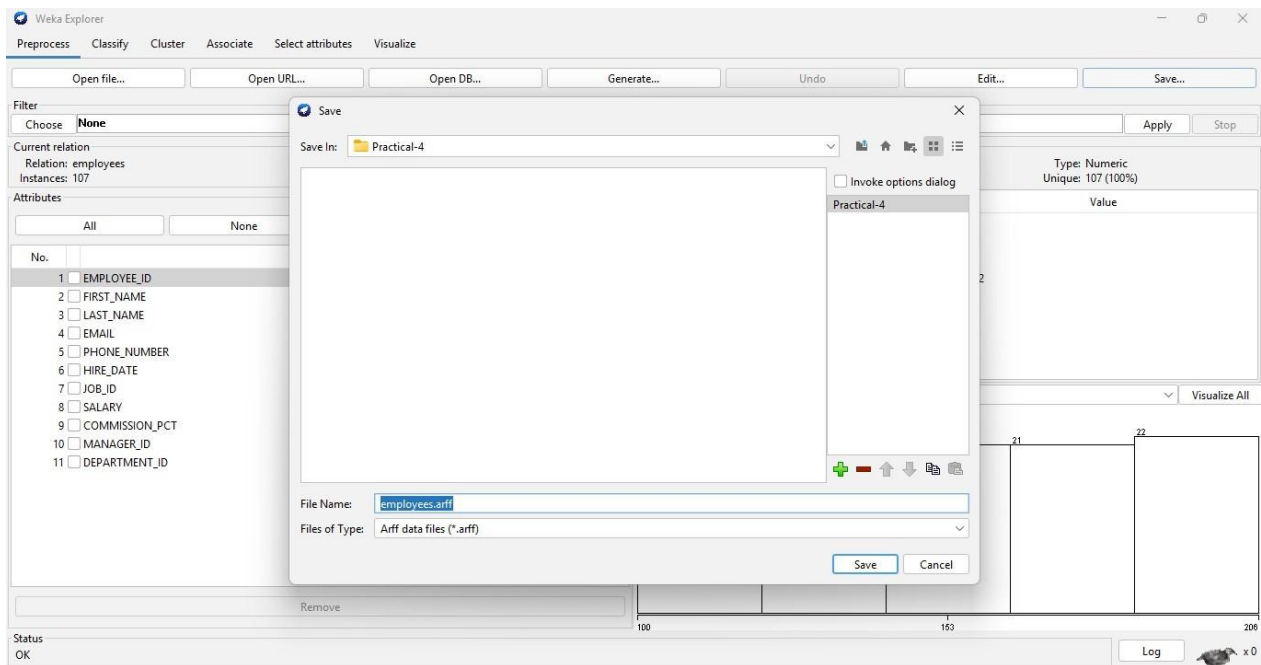
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Step-5: After loading the dataset, Click on **Save** then Choose file type as **ARFF**



Step-6: Converted into .arff

 employees	2/12/2026 7:49 PM	ARFF Data File	14 KB
 employees	2/1/2026 3:46 PM	Microsoft Excel C...	9 KB

Conclusion:

Weka is a powerful and user-friendly data mining tool used for classification, clustering, regression, and association rule mining. It supports multiple file formats, but ARFF is its default format. Using Weka Explorer, we can easily convert CSV files into ARFF format and perform various machine learning operations efficiently.